

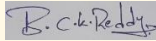
SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1023	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
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Date:	09/10/2026	Duration:	60 minutes

Code of Conduct

I B. Charan Kumar Reddy (192472195) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Answer

Agile Sprint Planning & User Story Workshop

Problem Statement: Autonomous Coding Agent Security and Software Supply-Chain Protection Platform

The proposed platform monitors AI coding agents, repositories, third-party dependencies, and CI/CD workflows to detect suspicious activities such as unauthorized actions, secret exposure, malicious dependency changes, and abnormal repository modifications. It supports risk assessment, human approval, security auditing, and isolation or rollback of suspicious changes.

1. Project Vision and Stakeholders

a) Project Vision

The vision of the project is to develop a secure and reliable platform that protects software development from AI-agent misuse and software supply-chain attacks. The platform will continuously monitor development activities, identify suspicious behavior, assess security risks, and prevent unauthorized or malicious changes from reaching production.

b) Project Objectives

- Monitor AI coding agent actions and repository modifications.
- Detect suspicious dependencies and malicious build activities.
- Identify secret exposure and unauthorized access.
- Maintain Software Bills of Materials (SBOMs) and audit trails.
- Integrate security checks into CI/CD pipelines.
- Provide risk scoring and human approval for sensitive actions.
- Support automatic isolation and rollback of suspicious changes.

c) Key Stakeholders and Responsibilities

Stakeholder	Responsibility
Developers	Develop code, use AI agents, and review changes.
Security Administrators	Define policies, monitor alerts, and investigate incidents.
DevOps Engineers	Manage CI/CD pipelines and integrate security checks.
Project Managers	Manage project progress, priorities, and resources.
AI Coding Agents	Generate and modify code and execute authorized tasks.
System Administrators	Manage permissions, infrastructure, and recovery.
Organization / Management	Define security requirements and evaluate risks.

2. Epics and User Stories

a) Epics

An epic is a large requirement that can be divided into smaller user stories.

Epic ID	Epic Name	Description
E1	AI Agent Monitoring	Monitor agent actions, commands, and file modifications.
E2	Dependency Security	Monitor third-party packages and maintain SBOMs.
E3	Repository Protection	Detect unauthorized and abnormal repository changes.
E4	CI/CD Security	Monitor build activities and pipeline configurations.
E5	Risk Management	Calculate risk scores and generate alerts.
E6	Incident Response	Support human approval, isolation, and rollback.
E7	Audit and Reporting	Maintain security records and generate reports.

b) User Stories

User stories are written using the Agile format:

As a user, I want a feature, so that I can achieve a benefit.

Story ID	User Story	Epic
US01	As a security administrator, I want to monitor AI-agent actions so that unauthorized activities can be detected.	E1
US02	As a developer, I want to track repository modifications so that unexpected code changes can be identified.	E3
US03	As a security administrator, I want to scan dependencies so that vulnerable or suspicious packages can be detected.	E2
US04	As a security administrator, I want to generate an SBOM so that all project dependencies can be tracked.	E2
US05	As a developer, I want to receive security alerts so that I can respond to suspicious activities quickly.	E5
US06	As a security administrator, I want to assign risk scores to security events so that critical threats receive priority.	E5
US07	As a security administrator, I want to approve sensitive agent actions so that dangerous operations are controlled.	E6
US08	As a DevOps engineer, I want to scan CI/CD workflows so that malicious build activities can be detected.	E4
US09	As a security administrator, I want to isolate suspicious changes so that they cannot affect other development activities.	E6

US10	As a developer, I want to roll back unsafe changes so that the repository can return to a trusted state.	E6
US11	As an auditor, I want to view security audit logs so that development activities can be investigated.	E7

3. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a user story to be accepted. The following criteria use the Given–When–Then format.

US01 – AI-Agent Monitoring

- Given: An AI coding agent is connected to the platform.
- When: The agent executes a command or modifies a file.
- Then: The platform records the action with its timestamp and relevant details.

US02 – Repository Monitoring

- Given: A repository is registered with the platform.
- When: A file or configuration is modified.
- Then: The platform records the change and flags modifications that violate configured security rules.

US03 – Dependency Scanning

- Given: A project contains third-party dependencies.
- When: A dependency scan is performed.
- Then: The platform identifies known vulnerabilities and reports packages that violate configured dependency policies.

US04 – SBOM Generation

- Given: A project has identifiable dependencies.
- When: The user requests an SBOM.
- Then: The platform generates an inventory containing the detected components and their available version information.

US05 – Security Alerts

- Given: A monitored activity violates a configured security rule.
- When: The platform detects the violation.
- Then: It generates an alert containing the event details and severity.

US06 – Risk Scoring

- Given: A security event has been detected.
- When: The risk assessment process evaluates the event.
- Then: The platform assigns a risk score based on configured factors and classifies the event according to defined thresholds.

US07 – Human Approval

- Given: An action requires approval under the security policy.
- When: An AI agent attempts that action.
- Then: The platform requests approval from an authorized person and prevents execution until approval is granted.

4. Product Backlog Prioritization

The product backlog contains all identified features, arranged according to business value, security importance, dependencies, and implementation needs.

Priority levels are High, Medium, and Low.

Priority	Story ID	Feature	Reason
High	US01	AI-agent monitoring	Provides visibility into agent activities.
High	US02	Repository monitoring	Detects unauthorized modifications.
High	US03	Dependency scanning	Identifies vulnerable dependencies.
High	US05	Security alerts	Enables timely responses to threats.
High	US06	Risk scoring	Helps prioritize critical security events.
High	US07	Human approval	Controls sensitive agent actions.
Medium	US04	SBOM generation	Improves dependency inventory and traceability.
Medium	US08	CI/CD security	Detects suspicious pipeline activities.
Medium	US09	Isolation	Limits the impact of suspicious changes.
Medium	US10	Rollback	Supports recovery from unsafe changes.
Medium	US11	Audit logs	Supports investigation and accountability.

Prioritization rationale: Monitoring, dependency scanning, and alerting are prioritized because they provide the initial security visibility needed by the platform. Risk assessment and approval build on this foundation. SBOM management, CI/CD security, and advanced incident-response features can then be implemented in subsequent iterations.

5. Sprint Backlog Planning

a) Sprint Duration

The first sprint is planned for two weeks.

b) Selected User Stories

The first sprint focuses on establishing basic monitoring and security detection.

Story ID	Selected User Story	Priority
US01	AI-agent monitoring	High
US02	Repository monitoring	High

US03	Dependency scanning	High
US05	Security alerts	High

c) Development Tasks

Story ID	Development Tasks
US01	Identify agent events; capture commands and file actions; store event details; test event recording.
US02	Connect a test repository; track file changes; identify configured policy violations; test change detection.
US03	Read dependency manifests; scan dependencies against an available vulnerability source; report findings; test scan results.
US05	Define alert severity levels; generate alerts from detected events; display event details; test alert generation.

d) Sprint Activities

- Week 1: Requirement clarification, task breakdown, initial monitoring implementation, and repository integration.
- Week 2: Dependency scanning, alert generation, integration testing, defect correction, and sprint review.

The selected stories are limited to foundational capabilities so the sprint remains achievable. Human approval, automatic rollback, and more advanced security functions can be planned for later sprints.

6. Story Point Estimation

Story points estimate the relative effort, complexity, and uncertainty of user stories. The Fibonacci sequence is used: 1, 2, 3, 5, 8, 13.

Story ID	User Story	Story Points	Justification
US01	AI-agent monitoring	5	Requires event capture, processing, storage, and testing.
US02	Repository monitoring	5	Requires repository integration and change detection.
US03	Dependency scanning	8	Requires dependency parsing and vulnerability analysis.
US05	Security alerts	3	Requires alert generation and severity display.

Total estimated sprint effort: 21 story points

Estimation Method

The team uses Planning Poker to compare story complexity and agree on relative estimates. The Fibonacci sequence reflects increasing uncertainty as tasks become more complex.

For example, generating a security alert is estimated at 3 points because it uses information from other components. Dependency scanning is estimated at 8 points because it requires dependency identification, vulnerability checks, and handling of scan results.

The 21-point commitment is a planning estimate, not a guarantee of completion. The team should confirm it against its actual capacity and previous sprint velocity.

7. Sprint Goal and Expected Deliverables

a) Sprint Goal

To establish the basic security monitoring capability of the platform by tracking AI-agent actions, detecting repository modifications, scanning third-party dependencies, and generating alerts for identified security issues.

b) Expected Deliverables

By the end of the first sprint, the team expects to deliver:

1. Basic AI-agent activity monitoring.
2. Repository modification tracking.
3. Dependency scanning with vulnerability findings from an available source.
4. Security alert generation for detected issues.
5. Integration tests for the selected user stories.
6. A sprint review report documenting completed work, test results, and unresolved issues.

c) Definition of Done

A user story is considered complete when:

- Its agreed acceptance criteria are satisfied.
- The implementation passes relevant tests.
- Identified defects are resolved or documented.
- The feature is integrated with the sprint build.
- The team has reviewed and accepted the work.

Conclusion

Agile sprint planning organizes the development of the Autonomous Coding Agent Security and Software Supply-Chain Protection Platform into manageable, prioritized tasks. Epics and user stories define the required functionality, acceptance criteria make requirements testable, and story points help estimate effort. The first sprint focuses on agent monitoring, repository protection, dependency scanning, and security alerts. These foundational capabilities prepare the platform for future work on risk-based approval, CI/CD security, isolation, and rollback.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25